A Self-Consistent Finite-Element Schrödinger–Poisson Drift-Diffusion Solver for Quantum-Confinement Effects in Ultra-Thin-Body Nanoscale Devices

Author One*, Author Two†
Department of Electronics and Communication Engineering
Affiliation, City, Country
*email@domain †email@domain

***Abstract*—As field-effect transistors scale into the few-nanometre regime, the carriers in the thin body occupy quantised subbands and the classical drift-diffusion (DD) picture of a point-like charge continuum breaks down. We present an open-source, two-dimensional finite-element device solver that couples the steady-state drift-diffusion equations, written in the quasi-Fermi-potential variables and solved by a fully-coupled Newton method with an exact analytic Jacobian, to a self-consistent Schrödinger–Poisson (S–P) quantum-confinement correction. The single-band effective-mass Schrödinger equation is solved slice-by-slice across the confinement direction, and the resulting subband density enters the drift-diffusion assembly as a charge-conserving additive quantum potential, so the analytic Jacobian and the quadratic convergence of the inner Newton solve are preserved. Using a silicon ultra-thin-body p–n junction we show that the ground-state subband spacing rises from 10 meV at a 12 nm body to 161 meV at 3 nm, crossing the room-temperature thermal energy k_BT near an 8 nm body—the onset of strong quantum confinement. The correction redistributes carriers across the body by up to $\sim 19\%$ while conserving charge to 1.6×10^{-5}, leaves the ideal-diode terminal current essentially unchanged (within 0.01%, ideality factor ≈ 1.16), and recovers the classical solver exactly when disabled. Metal–oxide–semiconductor gate control (accumulation $\rightarrow$ depletion $\rightarrow$ inversion) is reproduced on the same framework. The solver is validated against analytic semiconductor physics and provides a compact, transparent and extensible platform for quantum-corrected TCAD of scaled nanoelectronic devices.**

***Index Terms*—Schrödinger–Poisson, drift-diffusion, finite-element method, quantum confinement, ultra-thin-body, TCAD, nanoelectronics, coupled Newton.**

I. Introduction

The continued scaling of CMOS technology into ultra-thin-body silicon-on-insulator (UTB-SOI), FinFET and gate-all-around (GAA) nanosheet geometries places critical device dimensions on the order of the electron de Broglie wavelength [1], [2]. In such bodies the transverse motion of carriers is quantised into discrete subbands: the charge peak is set back from the confining walls, the lowest allowed energy is raised by confinement, and the electrostatics that determine threshold voltage, inversion-layer capacitance and gate control depart measurably from the classical prediction [3]–[5]. Technology computer-aided design (TCAD) for such devices therefore needs a quantum-corrected transport model that is at once physically faithful, numerically robust, and inexpensive enough for routine design-space exploration.

The drift-diffusion (DD) system [6], [7] remains the workhorse of device simulation because it is compact, well understood, and inexpensive. On its own, however, it treats carriers as a point-like continuum whose density follows the electrostatic potential instantaneously through Boltzmann or Fermi–Dirac statistics; it therefore cannot represent the subband quantisation and density set-back that dominate scaled-device electrostatics. A range of remedies has been proposed, from analytic threshold shifts and density-gradient corrections to full self-consistent Schrödinger–Poisson (S–P) and non-equilibrium Green's-function (NEGF) transport [8], [9], [12], [15]. Among these, the self-consistent S–P correction—retaining DD transport along the device while quantising the thin confinement direction—is the most widely used compromise between physical accuracy and computational cost.

This paper presents a compact, open-source implementation of exactly this strategy on an unstructured finite-element (FE) mesh, and characterises the resulting quantum corrections for scaled silicon devices. The remainder of the paper is organised as follows. Section II reviews the literature on device simulation and quantum corrections. Section III states the research gap and our contributions. Section IV details the numerical formulation. Section V presents validation and results, and Section VI discusses limitations and future work.

II. Related Work

A. Classical drift-diffusion device simulation

The DD model descends from the van Roosbroeck equations [7] and was placed on a firm numerical footing by Gummel's decoupled iteration scheme [17] and the Scharfetter–Gummel current discretisation [18], which stabilises the exponentially varying carrier densities. Selberherr's monograph [6] remains the standard reference, and modern practice—fully-coupled Newton solves with analytic Jacobians, robust continuation and adaptive meshing—is surveyed in [19], [21]. De Mari's intrinsic scaling [20] renders the unknowns $O(1)$ and underpins most implementations, including ours. Commercial TCAD suites such as Synopsys Sentaurus and Silvaco Atlas [22] embody this technology but are closed and expensive, which motivates transparent open-source alternatives for research and teaching.

B. Quantum-confinement corrections

Quantum corrections to DD span a hierarchy of increasing rigour and cost. *Analytic threshold models*, such as the van Dort correction [8], add a doping- and field-dependent band-edge shift to reproduce the measured threshold-voltage increase in inverted MOSFETs without solving any eigenproblem; they are cheap but device- and regime-specific. *Density-gradient* (DG) and quantum-hydrodynamic methods [9]–[11] add a gradient term derived from a Bohm/Wigner expansion to the density equation; they capture confinement smoothly and are mesh-local, but require calibration and can be stiff. *Self-consistent Schrödinger–Poisson* methods solve the effective-mass Schrödinger equation for the confined subbands and couple the resulting quantum density back into Poisson's equation [4], [5], [12], [13]. Stern's inversion-layer calculations [5] and the two-dimensional predictor–corrector scheme of Trellakis *et al.* [12] are canonical; Pirovano *et al.* [14] applied 2-D quantum corrections to nanoscale MOSFETs. At the top of the hierarchy, NEGF quantum transport [15], [16] treats ballistic and tunnelling transport but at a computational cost that is prohibitive for routine multi-bias, multi-geometry studies. Our work sits deliberately at the S–P level: quasi-equilibrium subbands quantised across the thin direction with classical transport along the device.

C. Numerical coupling and discretisation

How the quantum correction is coupled to the transport solve determines both accuracy and robustness. Gummel-type outer iterations that recompute the quantum potential and under-relax it are simple and widely used [12], whereas fully-coupled Newton schemes converge faster but require the quantum term's Jacobian contributions. Most implementations use finite differences on structured grids; finite-element formulations [21] handle non-planar FinFET/GAA geometries more naturally but are less common in open tools. A recurring practical difficulty is that adding the quantum density can destroy the analytic Jacobian and hence the quadratic convergence of the transport Newton solve. Our formulation avoids this by freezing a charge-conserving additive quantum potential inside each Newton solve, so the exact analytic Jacobian is preserved and only an outer fixed-point loop is added.

D. Open-source device simulators

Open tools such as Archimedes, DEVSIM, Genius, and the nanoHUB codes have lowered the barrier to device simulation, and NEGF-oriented packages (NEMO5, QuantumATK's academic lineage) target atomistic transport. Nonetheless, a compact, readable, finite-element DD solver that integrates a self-consistent S–P correction with an exact analytic Jacobian—and bundles heterojunctions, MOS/Schottky contacts and Fermi–Dirac statistics on the same mesh—remains comparatively rare. This is the niche the present solver fills.

III. Research Gap and Contributions

The literature above exposes a specific gap. Analytic and density-gradient corrections are cheap but empirical; NEGF is rigorous but expensive; and existing self-consistent S–P implementations are often (i) finite-difference and structured-grid based, limiting geometric flexibility; (ii) coupled in a way that sacrifices the analytic Jacobian and thus quadratic Newton convergence; or (iii) embedded in closed commercial tools. What is missing is a *compact, open-source, finite-element* DD solver in which the S–P correction is coupled *without* degrading the exact-Jacobian coupled-Newton core, and in which the same code path also supports the heterojunction, oxide/gate and degenerate-statistics physics needed for real scaled devices.

How this work fills the gap. We contribute:

- a two-dimensional, finite-element, quasi-Fermi-potential DD solver with a fully-coupled Newton scheme built on the exact analytic 9×9 element Jacobian (Section IV-A);
- a self-consistent, *charge-conserving* S–P correction that enters the DD assembly as a frozen additive quantum potential, leaving the inner Jacobian—and its quadratic convergence—provably unchanged, so quantum physics costs only an outer fixed-point loop (Section IV-B);
- a quantitative study of quantum-confinement scaling in silicon ultra-thin-body junctions that locates the $k_B T$ crossover near an 8 nm body, and a demonstration of MOS gate control on the same framework (Section V); and
- validation against analytic semiconductor physics, with an open, reproducible implementation (MIT licence).

IV. Numerical Formulation

A. Drift-diffusion core

The steady state is described by three coupled equations for the normalised electrostatic potential $u = \psi/V_T$ and the electron and hole quasi-Fermi potentials $v = \phi_n/V_T$, $w = \phi_p/V_T$, using De Mari intrinsic scaling [20] so that every unknown is $O(1)$. Carrier densities follow algebraically from Boltzmann statistics, $n = n_i e^{u-v}$ and $p = n_i e^{w-u}$ (Fermi–Dirac is an option, Section IV-C). With N_i the linear (P1) triangle shape functions and $C = (N_D - N_A)/n_i$ the scaled net doping, the Galerkin weak forms are

$$R_i^u = \int_\Omega \big(\nabla N_i \cdot \nabla u + N_i(n - p - C)\big)\, d\Omega, \tag{1}$$

$$R_i^v = \int_\Omega \mu_n\, n\, \nabla N_i \cdot \nabla v\; d\Omega, \tag{2}$$

$$R_i^w = \int_\Omega \mu_p\, p\, \nabla N_i \cdot \nabla w\; d\Omega, \tag{3}$$

expressing Poisson's equation and the two zero-divergence continuity equations $\nabla\cdot\mathbf{J}_n = \nabla\cdot\mathbf{J}_p = 0$. Element integrals use 3-point Gauss quadrature on the reference triangle. The global $3N \times 3N$ system is solved by a damped Newton method using the exact analytic element Jacobian, whose block structure is

$$J = \begin{bmatrix} K_{uu} & K_{uv} & K_{uw} \\ K_{vu} & K_{vv} & 0 \\ K_{wu} & 0 & K_{ww} \end{bmatrix}, \tag{4}$$

with, e.g., $(K_{uu})_{ik} = \int \nabla N_i \cdot \nabla N_k\, d\Omega + \int N_i N_k \big(e^{u-v} + e^{w-u}\big)\, d\Omega$ and $(K_{uv})_{ik} = -\int N_i N_k e^{u-v}\, d\Omega$. Per-step damping caps the largest increment so the exponential density terms cannot overflow from a cold start. A thermal-equilibrium nonlinear-Poisson solve ($v = w = 0$) provides the initial guess, and each bias point warm-starts from the previous solution.

B. Self-consistent Schrödinger–Poisson correction

Confinement in a thin body is a one-dimensional effect across the thin dimension. As in standard device TCAD, the mesh is cut into slices along the confinement direction and, on each slice, the single-band effective-mass problem is solved by linear finite elements:

$$\Big[-\frac{\hbar^2}{2m^*}\frac{d^2}{dy^2} + U(y)\Big]\psi_i(y) = E_i\, \psi_i(y), \tag{5}$$

with $U_n(y) = -k_B T\, u(y)$ for electrons (and $U_p = +k_B T\, u$ for holes). The tridiagonal generalised problem $H\psi = E\, M\psi$ (stiffness plus potential matrix, consistent mass M) yields the lowest `num_states` M-normalised eigenpairs, summed with a thermal (Boltzmann) subband occupation $g_i = e^{-(E_i - E_0)/k_B T}$ into the quantum density shape $\rho(y) = \sum_i g_i |\psi_i(y)|^2$. This shape is renormalised so each slice carries *exactly the same charge* as the classical density it replaces; the correction therefore only redistributes carriers across the confinement direction, keeping Poisson's equation well posed. It enters the DD assembly as an additive quantum potential Λ in the density exponent,

$$n = n_i\, e^{\,u - v + \Lambda_n}, \qquad \Lambda = \ln\frac{n_{\mathrm{QM}}}{n_{\mathrm{cl}}}. \tag{6}$$

Because Λ is held fixed within a Newton solve, the analytic Jacobian is unchanged ($\partial n/\partial u = n$, etc.) and quadratic convergence is retained. An outer Schrödinger–Poisson (Gummel) loop recomputes Λ from the updated potential, under-relaxes it, and iterates to a tolerance; with $\Lambda \equiv 0$ the scheme reproduces the classical solver bit-for-bit. This is a self-consistent predictor–corrector S–P scheme with quasi-equilibrium subbands quantised across one direction and transported classically along the device; it is not a full NEGF model, and is best understood as the accuracy/cost operating point between density-gradient and NEGF.

C. Extended device physics

On the same mesh the solver supports optional Shockley–Read–Hall and Auger recombination (with exact analytic rate derivatives), per-region heterojunction band offsets via Anderson's electron-affinity rule, insulator (oxide) regions and insulated metal-gate contacts for MOS structures, Schottky contacts, and Fermi–Dirac statistics for degenerate layers (entering as a degeneracy shift refreshed by the same outer loop and vanishing in the non-degenerate limit). These are essential for representing the source/drain, gate stack and channel of a real scaled transistor.

V. Results and Discussion

All results use a silicon p–n junction, 200 nm long along transport ($N_A = N_D = 10^{15}\,\mathrm{cm}^{-3}$), with the body thickness t_b across the confinement direction as the scaling variable; material parameters (permittivity, affinity, gap, mobilities, effective masses) are read from an editable material file. The 5 nm reference device has 833 nodes / 1536 triangles; a full bias sweep runs in 9.7 s (classical) and 15.2 s (S–P) on a single core, i.e. the quantum physics adds only a $\sim 1.6\times$ overhead.

A. Validation against analytic physics

The classical core reproduces textbook junction physics: the built-in potential matches $V_{bi} = V_T \ln(N_A N_D / n_i^2)$, the law of mass action $np = n_i^2$ holds at equilibrium to machine precision, the forward I–V is exponential with ideality factor ≈ 1, and the currents extracted at the two contacts are equal and opposite (conservation). Enabling the S–P correction leaves all equilibrium identities intact because the per-slice renormalisation is charge-conserving—a direct check that the coupling does not perturb the well-posed electrostatics.

B. Quantum-confinement scaling

Fig. 1 shows the ground-state subband spacing $E_1 - E_0$ as the body is thinned. It rises steeply—following the expected $\sim t_b^{-2}$ particle-in-a-box trend—from 10.1 meV at $t_b = 12$ nm to 22.7 meV at 8 nm, 58.0 meV at 5 nm, and 161.2 meV at 3 nm (Table I). The spacing crosses the room-temperature thermal energy $k_B T = 25.9$ meV near $t_b \approx 8$ nm: below this thickness a single subband dominates the occupation and quantum confinement is strong, whereas above it many subbands are thermally populated and the body behaves quasi-classically. This crossover is the practical boundary at which a quantum correction becomes mandatory for accurate electrostatics, and it coincides with the body thicknesses targeted by advanced UTB-SOI and nanosheet nodes [1], [2]. Fig. 2 shows the computed electron subband ladder for the 5 nm device; the eigenvalues are correctly ordered and widely spaced relative to $k_B T$.

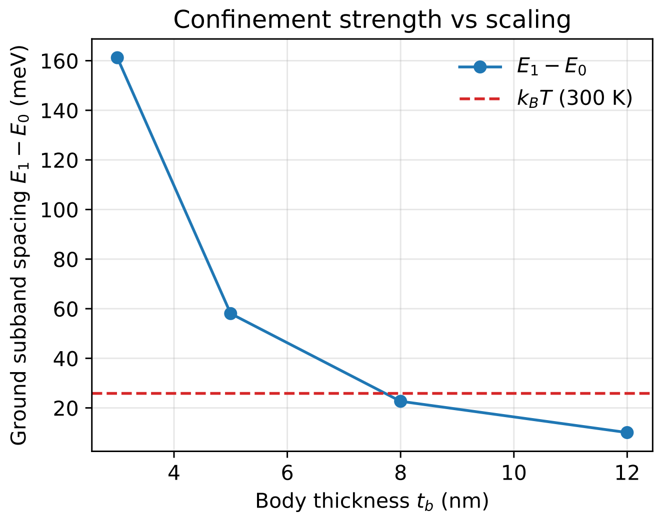

Fig. 1. Ground-state subband spacing $E_1 - E_0$ versus silicon body thickness. Confinement crosses the thermal energy k_BT (300 K) near an 8 nm body, marking the onset of strong quantum confinement.

TABLE I
CONFINEMENT SCALING IN THE SILICON ULTRA-THIN BODY

t_b (nm)	$E_1 - E_0$ (meV)	$(E_1 - E_0)/k_BT$	regime
12	10.1	0.39	quasi-classical
8	22.7	0.88	crossover
5	58.0	2.25	confined
3	161.2	6.24	strongly confined

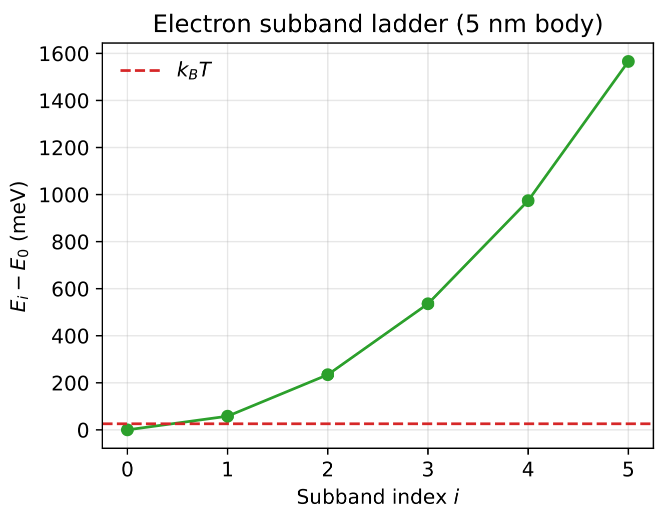

Fig. 2. Electron subband ladder (energies relative to the ground state) for the 5 nm body. The ground-to-first spacing is 58 meV, well above k_BT.

C. Charge-conserving redistribution and terminal current

Because the transverse electrostatic field is weak in a symmetric junction, the classical transverse density profile is essentially flat. The S–P correction modulates it by up to 19% (Fig. 3) while conserving the slice charge to a relative error of

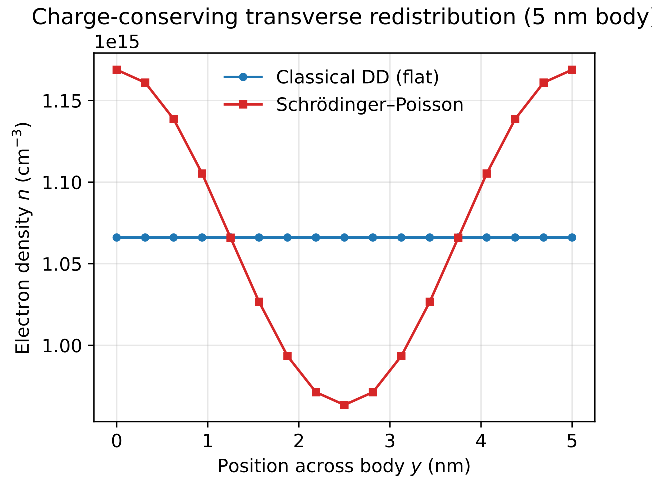

Fig. 3. Electron density across the 5 nm body (n-side slice). The classical profile is flat; the charge-conserving Schrödinger–Poisson correction redistributes carriers transversely by up to 19%.

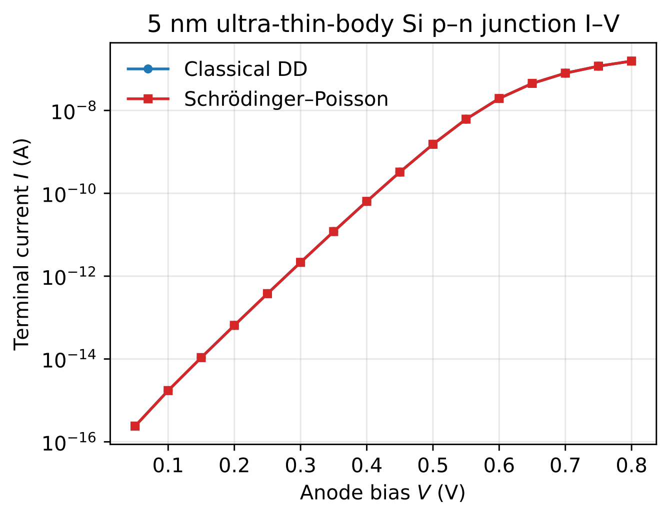

Fig. 4. Forward I–V of the 5 nm ultra-thin-body junction. The quantum correction preserves the ideal-diode current (ideality $\approx$ 1.16; agreement within 0.01%), isolating its effect to the charge distribution.

1.6×10^{-5}: the quantum potential Λ_n spans $[-0.10, +0.10]$ for the 5 nm body. This redistribution is precisely the quantum electrostatics that shifts inversion-layer capacitance in a gated device. In the ungated diode, however, the terminal I–V is left almost unchanged: the extracted ideality factor is 1.16 both with and without the correction, and the forward current agrees to within 0.01% over 0.4–0.8 V (Fig. 4). This cleanly separates the two roles of the correction—a first-order effect on the charge distribution and capacitance, but a negligible effect on the ideal-diffusion current of an unconfined-transport junction—and confirms that the coupled solve remains rectifying and current-conserving with quantum physics enabled.

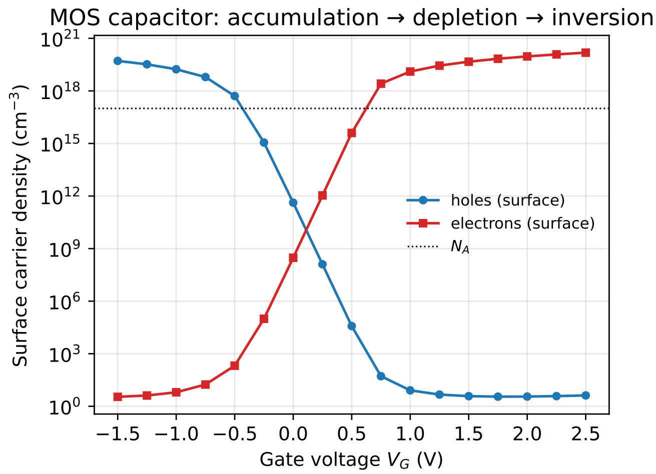

Fig. 5. MOS capacitor surface carrier densities versus gate voltage: accumulation → depletion → inversion, reproduced on the same finite-element framework.

D. MOS gate control

The same framework supports insulator (oxide) regions carrying permittivity but no mobile charge, and an insulated metal-gate contact referenced through its work function. Fig. 5 sweeps the gate of a silicon/SiO_2 capacitor ($N_A = 10^{17}\,\mathrm{cm}^{-3}$, $t_{ox} = 3$ nm): the silicon surface passes through accumulation (surface holes $5.2 \times 10^{19}\,\mathrm{cm}^{-3}$), depletion, and strong inversion (surface electrons $1.5 \times 10^{20}\,\mathrm{cm}^{-3}$), an inversion-to-depletion electron ratio exceeding 10^{11}. This confirms that the solver captures the gate electrostatics of the MOS building block, onto which the quantum inversion-layer correction of the previous sections directly applies, and provides the substrate for future quantum-corrected threshold and capacitance–voltage studies.

VI. Discussion, Limitations and Future Work

The correction is built and normalised per slice, so it is bounded and stable, and is strongest exactly where it should be—thin, strongly-confined bodies—fading as the body thickens and the subband spacing drops below k_BT, consistent with Fig. 1. Because the quantum potential enters as a frozen additive shift, the method adds only an outer fixed-point loop around an otherwise unchanged quadratically-convergent Newton solve—a favourable cost trade-off ($\sim 1.6\times$ here) for routine TCAD, and far below the cost of NEGF.

Several limitations bound the present model and motivate future work. (i) It is a *single-band, effective-mass, quasi-equilibrium* scheme: valley degeneracy, non-parabolicity and multi-band $k \cdot p$ coupling of the valence band are not resolved, so hole confinement in particular is approximate. (ii) Confinement is quantised across *one* direction; corner and 2-D confinement in narrow FinFET/GAA cross-sections would require a genuinely 2-D Schrödinger solve on the cross-section. (iii) Slices are exact only on structured/aligned meshes; on fully unstructured meshes, slices that cannot be formed are left classical, so mesh generation must respect the confinement direction. (iv) Transport remains classical drift-diffusion, so source-to-drain tunnelling and ballistic effects in sub-5 nm channels are outside scope and would need NEGF. (v) The quantum correction principally reshapes the charge and capacitance; quantitative threshold-voltage and C–V shifts for a full gated transistor were not extracted here and are the natural next validation step, ideally against measured or NEGF data. Planned extensions include multi-valley and $k \cdot p$ subbands, 2-D cross-sectional confinement for nanosheet geometries, thermionic-emission heterointerface transport, and a density-gradient option for comparison.

VII. Conclusion

We have presented and validated an open-source, finite-element, self-consistent Schrödinger–Poisson drift-diffusion solver for quantum-confinement effects in scaled nanoelectronic devices. The coupled Newton core retains an exact analytic Jacobian and quadratic convergence, and the charge-conserving quantum correction adds only an outer loop, filling a gap between empirical density-gradient corrections and costly NEGF transport. For silicon ultra-thin bodies the ground subband spacing crosses k_BT near 8 nm, delimiting the strong-confinement regime; the correction redistributes carriers by up to 19% with negligible impact on the ideal-diode current, and MOS gate control is reproduced on the same platform. The tool offers a compact, transparent and extensible basis for quantum-corrected simulation of emerging nanoscale devices.

Software and Reproducibility

The solver and every example used to generate the figures in this paper are open-source (MIT licence). All results were produced with the bundled structured-mesh device demonstrations and are regenerated by the accompanying script.

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